



1. Binocular Rivalry

Visual perception *alternates* between different images presented simultaneously to each eye. Characterised by:

- Perceptual Dominance
- Suppression
- Stochastic Switching

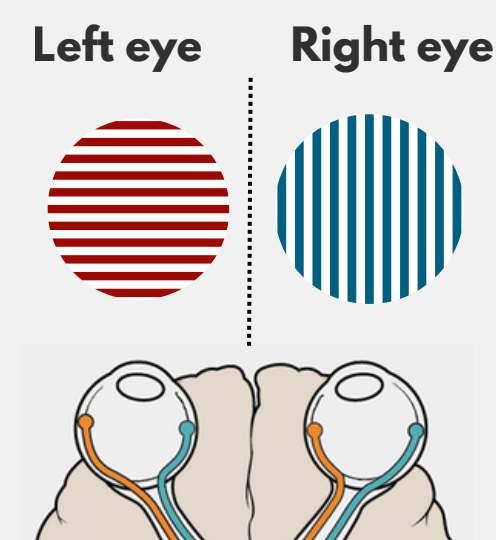


Fig 1. Binocular Rivalry Illustration

2. Minimal Rivalry Model

Mutual inhibition between monocular populations (E_L, E_R) and slow self-adaptation generate a stable limit cycle causing alternating perceptual dominance.

$[]_+$ represents a threshold nonlinearity that gives a 0 output for a negative input.

$$\begin{aligned}\tau \frac{dE_L}{dt} &= -E_L + M[L(t) - aE_R + \epsilon E_L - gH_L]_+ \\ \tau_H \frac{dH_L}{dt} &= -H_L + E_L \\ \tau \frac{dE_R}{dt} &= -E_R + M[R(t) - aE_L + \epsilon E_R - gH_R]_+ \\ \tau_H \frac{dH_R}{dt} &= -H_R + E_R\end{aligned}$$

3. Simulations of Levelt's 2nd Law

Reducing contrast in one eye increases dominance duration in the other eye without affecting its own duration.

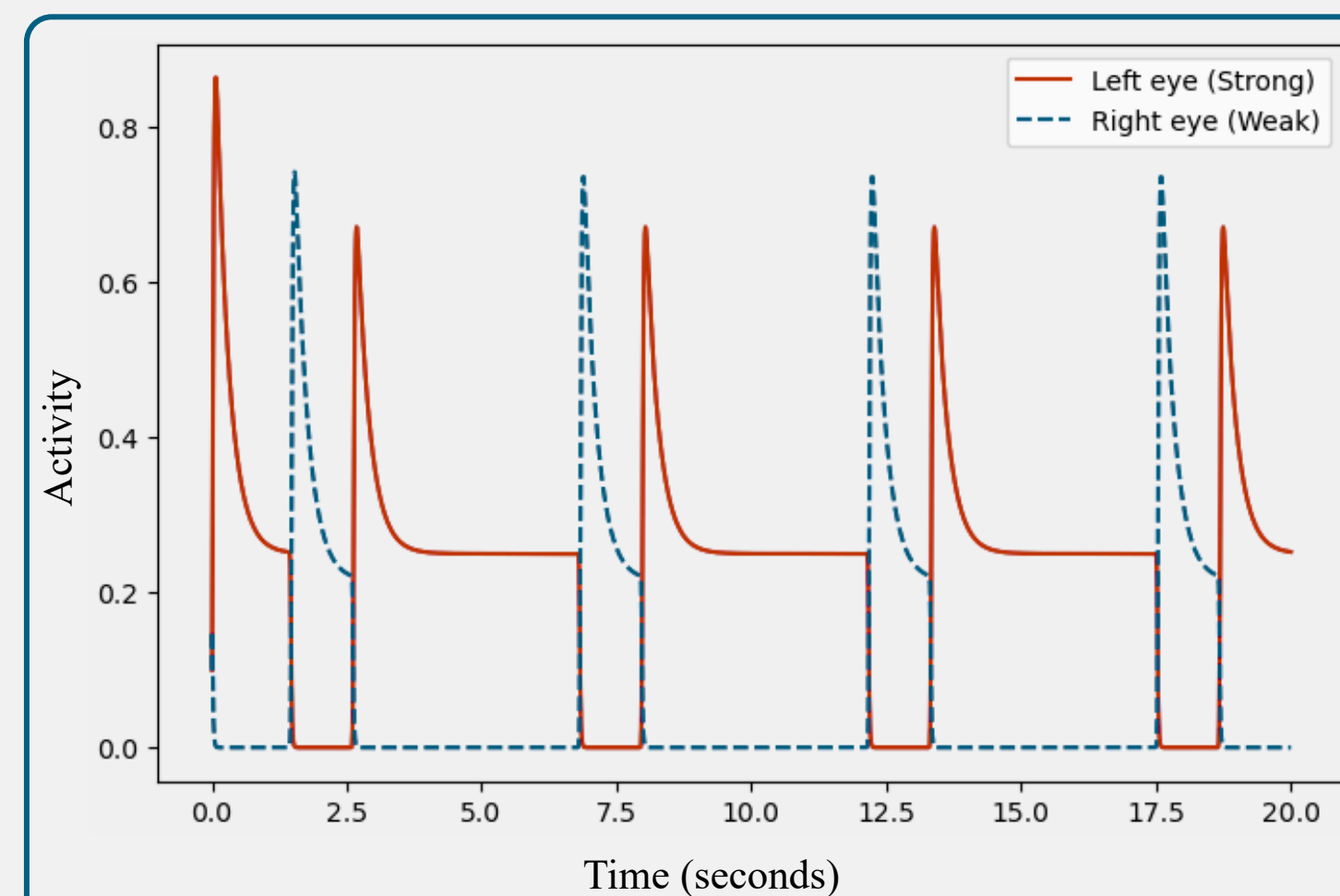


Fig 2. Larger dominance duration for the left eye when the stimulus strength of the right eye is decreased.

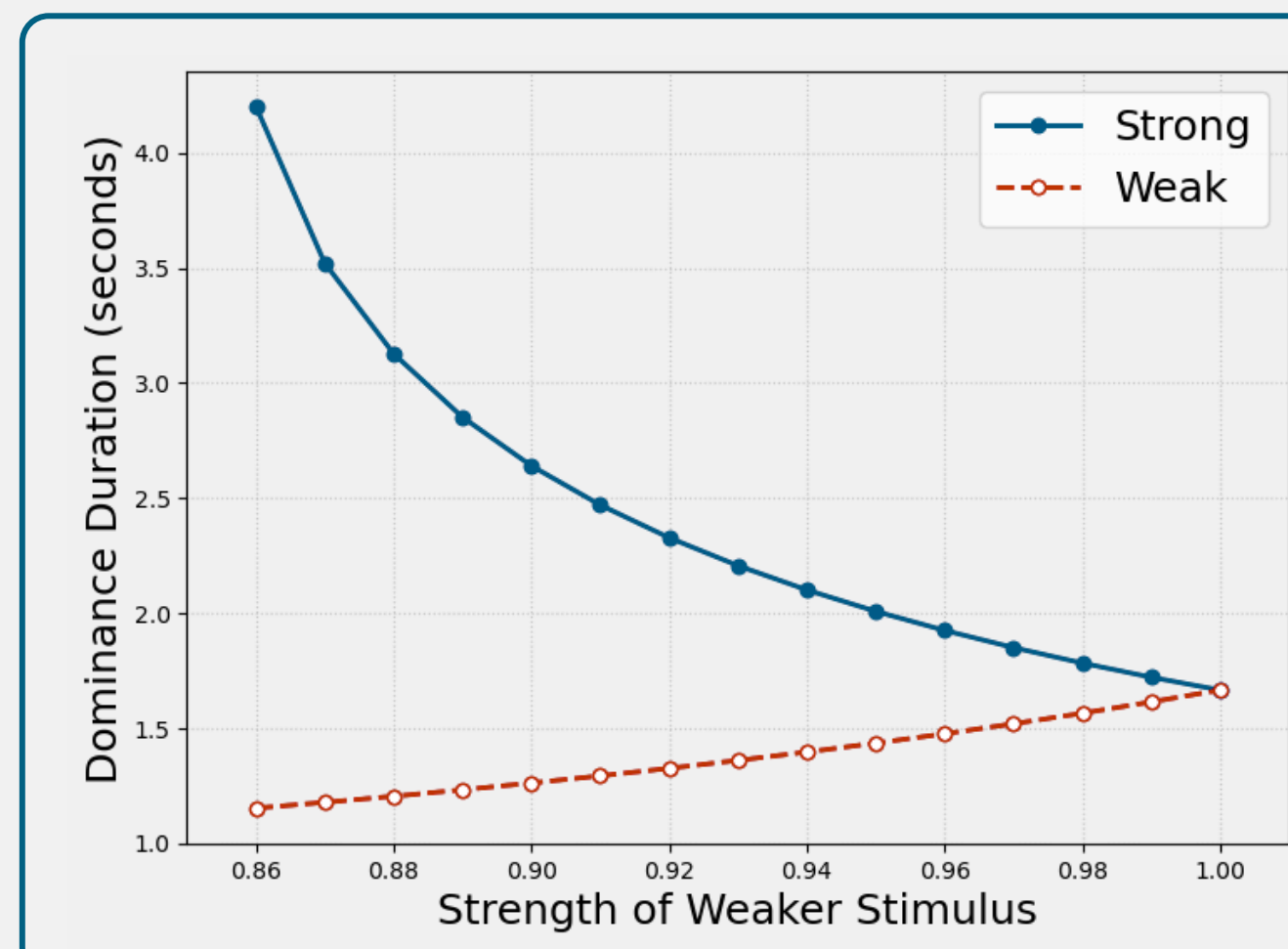


Fig 3. When strength < 0.85, the system undergoes a bifurcation and transitions from a limit cycle to WTA.

4. Dominance Durations w.r.t Model Parameters

For symmetric initial conditions ($L=R$):

The bifurcation to WTA occurs when: $g = a - \frac{1}{M} + \epsilon$, and

$T = \tau_H \ln \left(\frac{g}{1/M + g - \epsilon - a} \right)$ gives the dominance duration.

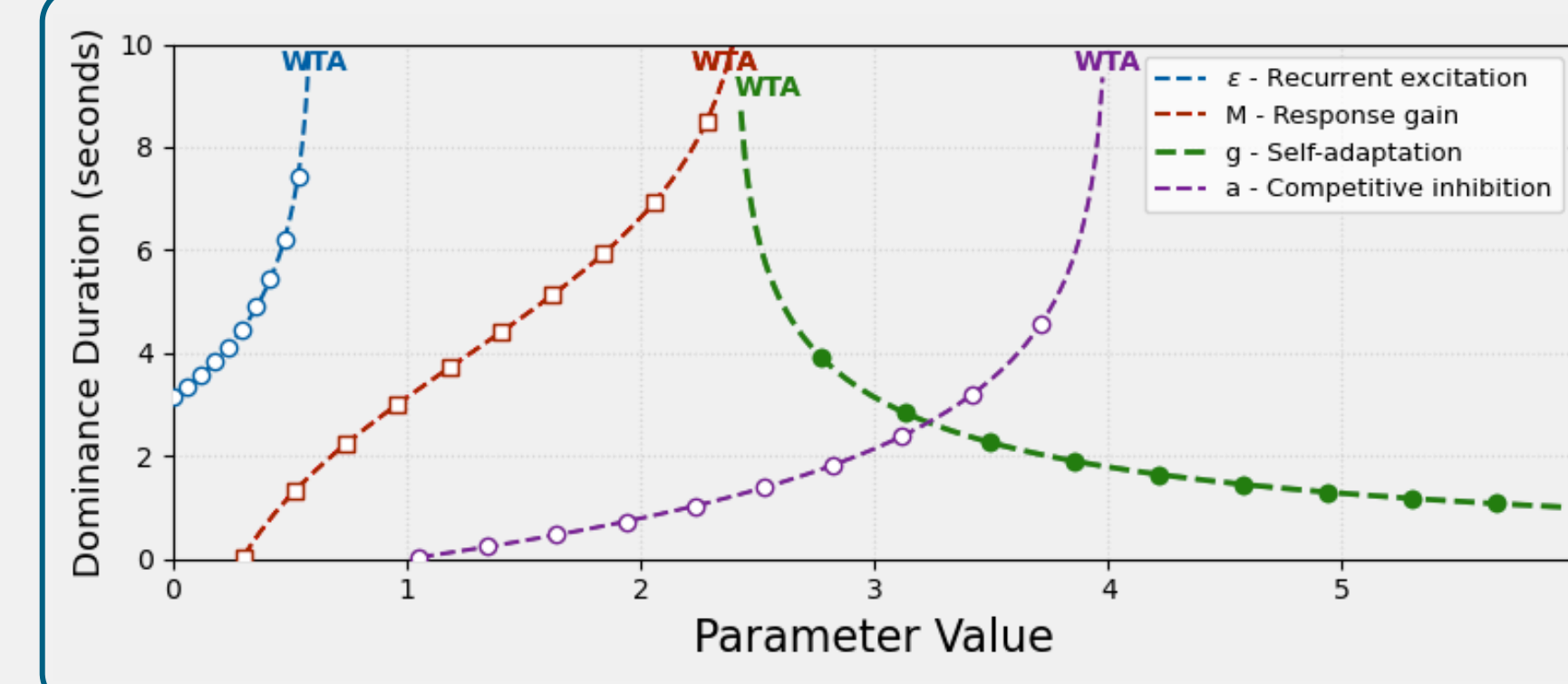


Fig 4. WTA represents the point at which rivalry stops and current dominant eye remains dominant indefinitely.

5. Fast-Scale Stability Analysis

When adaptation (H) is much slower than neural activity ($\tau_H \gg \tau$), it can be treated as a constant for short-term stability analysis (H was fixed at 0).

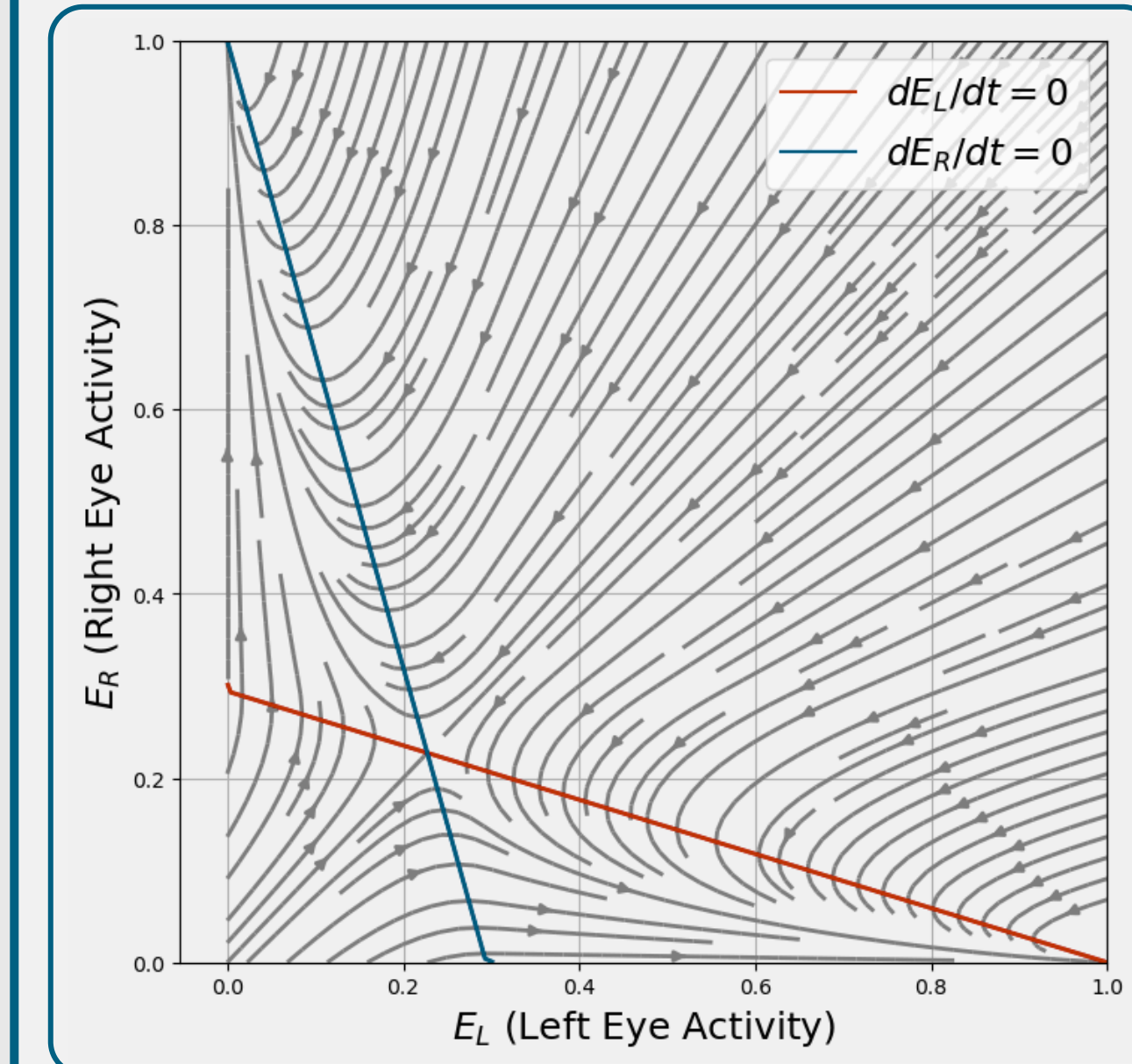


Fig 5. Phase plane of a reduced system where H is treated as constant. The instability of the intersection confirms the WTA dynamic - dominance in one eye.

6. Bifurcation Analysis - 4D System

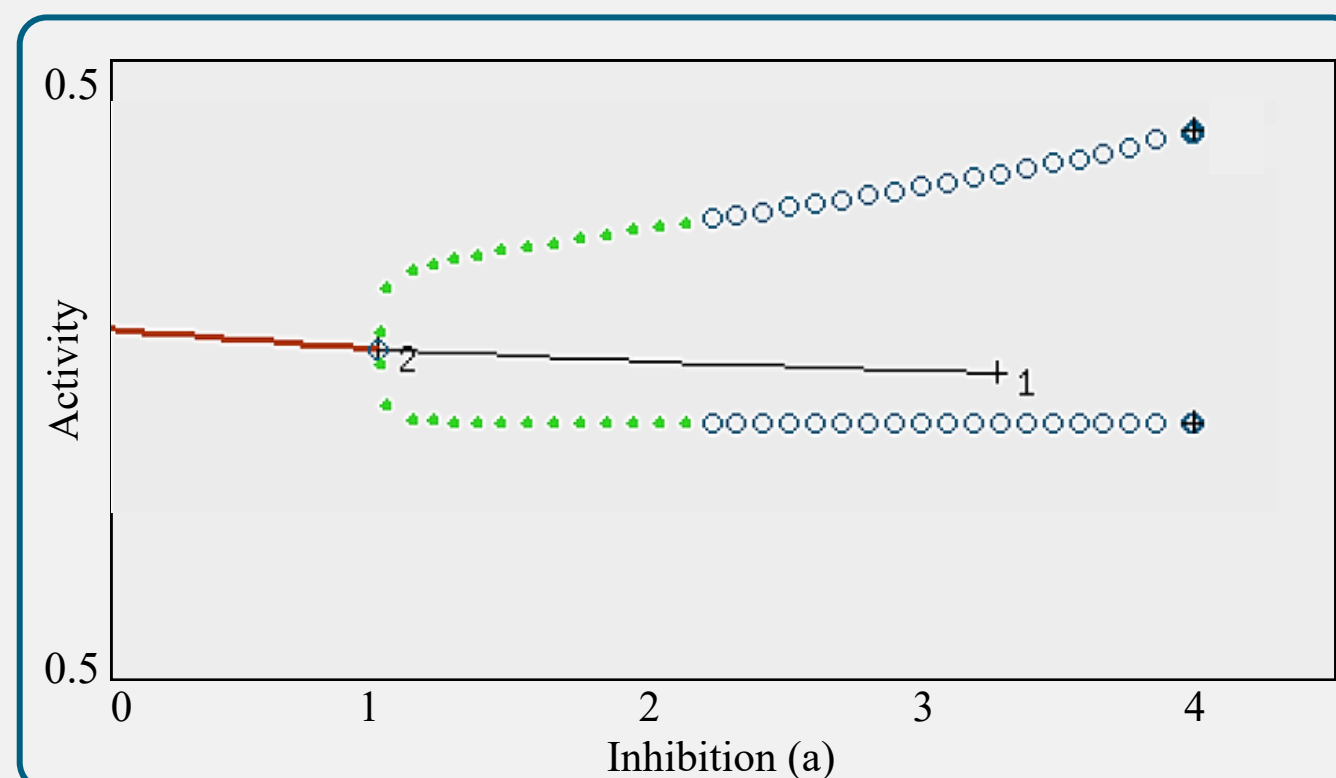


Fig 6. Increasing inhibition triggers a Hopf bifurcation at $a = 1.04$, marking the transition into sustained binocular rivalry oscillations (limit cycles).

7. Perceptual Memory

The same eye remains dominant after the stimulus is temporarily removed.

- Stimulus-off period: neural activity (E) drops to 0 but the Facilitation (F) decays slowly.
- Stimulus reappearance: the previously dominant eye possesses a *facilitated* recurrent gain ($\epsilon + F$)

$$\begin{aligned}\tau \frac{dE_L}{dt} &= -E_L + M[L(t) - aE_R + (\epsilon + F_L)E_L - gH_L]_+ \\ \tau_F \frac{dF_L}{dt} &= -F_L + fE_L\end{aligned}$$

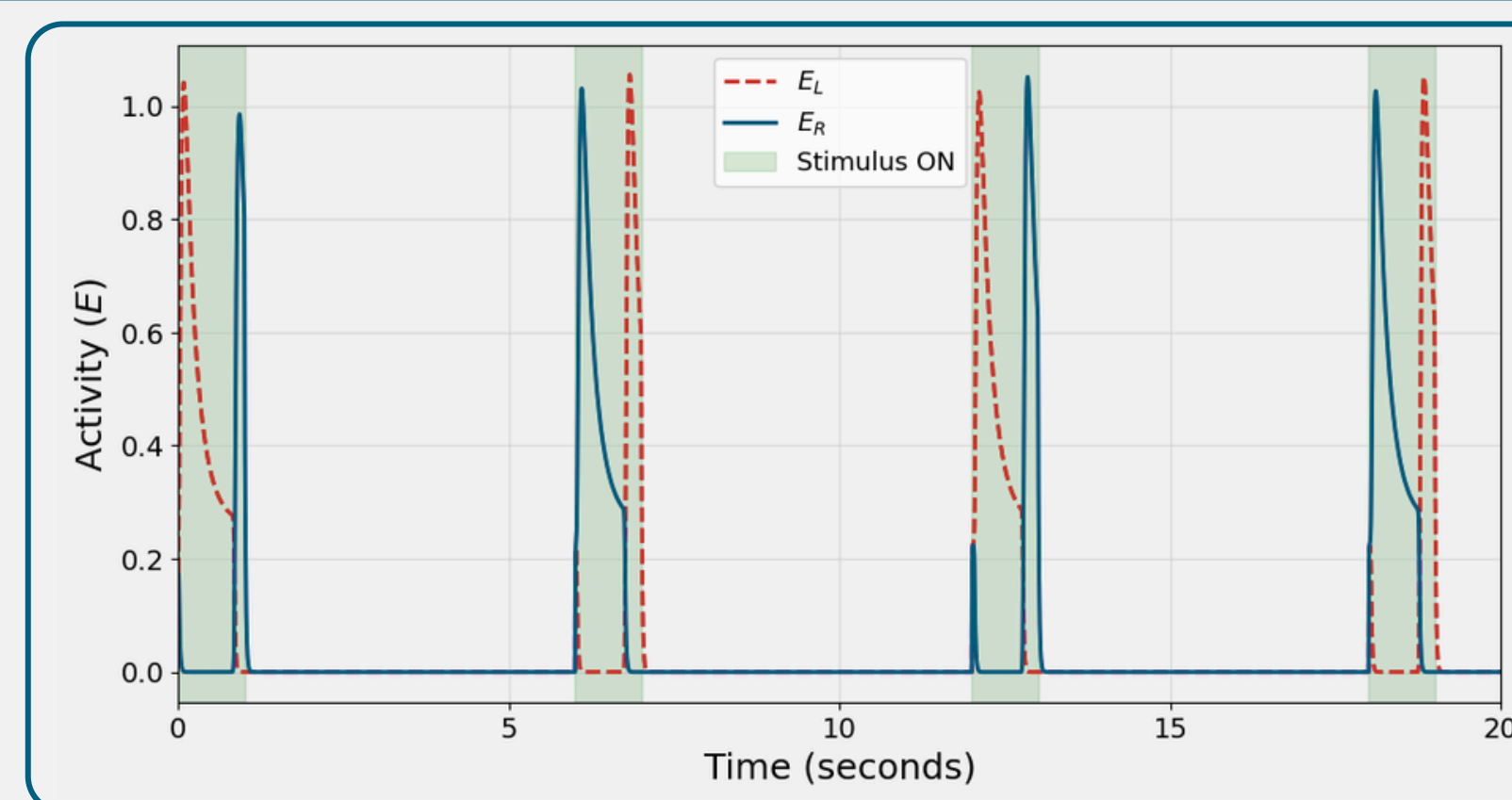


Fig 7. Simulation of a periodic 1s ON / 5s OFF sequence. Memory is maintained by slow recurrent facilitation (F). Successive reductions in dominance durations is due to the recovery period being insufficient for H to return to a baseline state given the time constant $\tau_H \approx 0.996s$

Instability/WTA Condition

Stability thresholds:

- Limit cycle: $\frac{1}{M} - \epsilon < a < \frac{1}{M} - \epsilon + g \frac{R}{L}$
- Fusion (stability): $a < \frac{1}{M} - \epsilon$
- WTA: $a > \frac{1}{M} - \epsilon + g \frac{R}{L}$

$$J = \frac{1}{\tau} \begin{pmatrix} M\epsilon - 1 & -Ma \\ -Ma & M\epsilon - 1 \end{pmatrix}$$

The instability is determined by the leading eigenvalue, $\lambda_{\max} > 0$, indicating an unstable steady state.

References

- Chen, X., & He, S. (2004). Local factors determine the stabilization of monocular ambiguous and binocular rivalry stimuli. *Current Biology*, 14, 1013–1017.
- Wade, N. J. (1998). *A natural history of vision*. Cambridge, Mass: MIT Press.
- Wilson, H. R. (2007). Minimal physiological conditions for binocular rivalry and rivalry memory. *Vision Research*, 47(21), 2741–2750.
- <https://doi.org/10.1016/j.visres.2007.07.007>